[METRIC]
H-B-711K
July 6, 1995
Superseding
H-B-711J
August 25, 1993

FEDERAL SPECIFICATION

BRUSHES, VARNISH, OVAL

1. Scope and Classification.

1.1 This specification covers oval, metal-bound, varnish brushes made with natural hog bristle or synthetic filament. These brushes are designed and manufactured for painting small, narrow, or corner areas.

1.2 Classification. The brushes shall be of the following types and sizes as specified (see 6.2): (Size denotes the smaller diameter inside the ferrule).

Type I -Bristle
Type II -Synthetic

Size, mm, (inch)	32 (1-1/4)
	38 (1-1/2)

2. Applicable Documents.

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for approval, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-1880	-Brushes; Preparation for Delivery of
H-S-951	-Synthetic Filaments for Brushes, General Spec For

(Copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available from the following address: GSA Specifications Section (3FBP-W), Suite 8100, 470 L'enfant Plaza SW, Washington, DC 20407.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 -Sampling Procedures and Tables for Inspection by attributes

Beneficial comments, recommendations, additions, deletions, clarification, etc., and any data which may improve this document, should be sent to: GSA, Paints and Chemicals Commodity Center, 400 15th Street, S.W., Auburn, WA 98001.

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(Copies of Federal Specifications, Standards, and Regulations required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

American Society for Testing and Materials:

ASTM D13 -Specification for Spirits of Turpentine

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

3. Requirements.

3.1 Materials and components. Materials and components shall conform to the requirements specified herein.

3.1.1 Bristles (Type I). The bristles for Type I brushes shall be unbleached and undyed natural hog bristles. The bristles shall be boiled or steamed and straightened so that it will not twist or curl, and shall be free from reconditioned bristle or adulterants. The bristles shall pass the straightness test specified in 4.4.4. A certificate of compliance from the supplier that the bristle meets all requirements (stiffness and sterilization) may be accepted. The contractor shall provide certification that color(s) has not been added to the bristles.

3.1.2 Synthetic filaments (Type II). The synthetic filaments for Type II brushes shall be an organic polymeric material conforming to H-S-951, Class N, type I, or Class E, type I, tapered filaments meeting the size designation of Table I. All filaments shall be tipped to a depth of not less than 5 mm (3/16 inch). Twenty percent, by weight of the top-sized filaments, shall have split or flag ends to better hold and distribute paint uniformly.

3.1.3 Ferrule. The ferrule shall be made of corrosion resistant metal, or metal made corrosion resistant by electro-plating.

3.1.4 Handle. The handle shall be smoothly finished close grained hardwood or a synthetic material with all sprues and runners removed. The synthetic handle shall not crack or break when tested as in 4.4.3, and shall show no perceptible change when tested as in 4.4.2.1.

3.1.5 Filler strips. Filler strips shall be made of wood, corrosion resistant metal, or plastic, and shall not soften or otherwise change when tested as in 4.4.2.2.

3.1.6 Setting compound. The setting compound shall be a material that shall withstand the test specified in 4.4.2 without softening or deteriorating, and without loosening of bristle or filament in the compound.

3.2 Construction and physical requirements.

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3.2.1 Ferrule. The ferrule metal shall have a minimum wall thickness of 0.25 mm (0.010 inch). The ferrule height shall be 25.4 mm (1-inch) +/- 1.0 mm (0.039 inch).

3.2.2 Set. The butt ends of the bristles or filaments shall be set a minimum of 6 mm (1/4 inch) in a setting compound conforming to 3.1.6. When tested, as in 4.4.2, there shall be no more than a total of five bristles or filaments pulled loose during the tensile pull test.

3.2.3 Assembly. The ferrule shall be fastened to the handle with a minimum of three 17 gauge, or larger, corrosion resistant nails. The nails shall be 10 mm (3/8 inch) minimum in length. The nails shall be evenly spaced around the ferrule.

3.2.4 Trim. Brushes shall be cupped or machine chiseled with a minimum chisel of 6 mm (1/4 inch).

3.3 Composition and physical requirements.

3.3.1 Type I. The bristle shall be 70 percent top size and blended in accordance with Table I. (Dressed bristle, described as 70 percent top size, denotes that bundle consists of 70 percent bristle measuring 1.5 mm (1/16 inch) or more above the length specified to 3 mm (1/8 inch) below the length specified, but including a few unavoidable shorts).

3.3.2 Type II. Synthetic filaments conforming to 3.1.2 shall be blended as specified in Table I, with a tolerance of plus or minus 3 percent for each length and stiffness specified.

Table I. Physical requirements for Types I and II brushes

Characteristic	Brush size, mm, (inches) 32 (1-1/4)	38 (1-1/2)
Ferrule		
Smaller inside diameter, mm, (inches), min	31 (1-7/32)	37 (1-15/32)
Greater inside diameter mm, (inches), min.	40 (1-19/32)	47 (1-27/32)
Bristle (Type I)		
Weight of bristle per brush, grams, min.	40	58
Length outside of ferrule, mm, (inches), min.	79 (3-1/8)	92 (3-5/8)
Filaments (Type II)		
Weight of filaments per brush, grams, min.	35	58

Table I. Physical requirements for Types I and II brushes (continued)

Characteristic	Brush size, mm, (inches) 32 (1-1/4)	38 (1-1/2)

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Blend of filaments, percentage and length, mm, (inches), min.		
Size 12-8	60% 89 (3-1/2) 15% 83 (3-1/4) 15% 76 (3) 10% 70 (2-3/4)	60% 108 (4) 15% 95 (3-3/4) — —
Size 15-10	— —	15% 89 (3-1/2) 10% 83 (3-1/4)
Length outside ferrule,	78 (3-1/16)	91 (3-9/16)

3.4 Marking for identification. Unless otherwise specified (6.2), and in addition to any special marking required by the contract or order, each brush shall be stamped or otherwise permanently marked with the manufacturer's name or trademark and the National Stock Number (NSN).

3.5 Workmanship. The finished brushes shall be clean, well made, and free from any defect which may affect serviceability or appearance. There shall be no sharp edges, splinters, burrs, or other imperfections. The brushing part shall be trimmed evenly and uniformly with the flagged ends preserved to the finished length outside the ferrule.

4. Quality Assurance Provisions.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements specified herein using facilities acceptable to the Government. The Government reserves the right to perform any inspection herein when deemed necessary to assure that the brushes conform to prescribed requirements.

4.2 Inspection of preparation for delivery. An examination shall be made to determine compliance with 3.3 and section 5 of this specification. The examination shall be conducted as specified in PPP-B-1880.

4.3 Inspection of the end item. Lots shall be assembled and randomly sampled in accordance with MIL-STD-105, inspection level II, with an acceptable quality level (AQL) of 2.5 percent defective. Sample brushes shall be examined for the following attributes:

Bristles/filaments
Ferrules
Handles
Construction
Set
Trim
Workmanship
Identification markings

Any brush in the sample containing one or more visual or dimensional defects shall be considered a defective brush.

4.3.1 Sampling for testing of the end item. A random sample of brushes shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level (AQL) of 2.5 percent defective.

4.4 Testing procedures. Samples, selected in accordance with 4.3.1, shall be subjected to the following tests. Failure to pass any test shall be cause for rejection of the lot. All tests shall be evaluated for conformance to the requirements in section 3.

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4.4.1 Weight of bristle or filament. After removal of the ferrule, the setting compound shall be removed from the bristle or filament by any practical means which does not injure the bristle or filament. The weight of bristles or filaments shall be determined for compliance with physical requirements.

4.4.2 Solvent resistance. Solvent resistance tests shall be performed at room temperature.

4.4.2.1 Handle (Organic coating). Brush handles with organic coating shall be tested as follows:

A. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in a commercially available mineral spirits. After 30 minutes, the handle shall be removed and examined. There shall be no wrinkling, blistering, or separation of the coating.

B. With the brushing end up, approximately one-half of the handle ~~shall be immersed~~ for 30 minutes in turpentine conforming to ASTM D13. After 30 minutes, the handle shall be removed from the turpentine and examined. There shall be no wrinkling, blistering, or separation of the coating.

C. The brush handle shall be dried for 2 hours and the organic coating shall be examined. The coating shall not be wrinkled, blistered, or separated from the handle and the coating shall not be soft.

4.4.2.2 Base material, filler strips, synthetic handles. The brush shall be tested for solvent resistance using a commercially available lacquer thinner. (NOTE: The organic coating on the handle may be removed by this solvent. The objective is to test solvent resistance of the handle base material, filler strips, and the setting compound.)

The brush shall be suspended so that the brushing part and approximately one-half of the ferrule are immersed in commercial lacquer thinner for 1 hour. Synthetic handles shall be immersed to a depth of one-half of the handle length. The brush shall be removed and examined as follows:

A. Handle. As soon as the brush is removed from the lacquer thinner, the brush shall be dried for 2 hours and examined for deterioration of the handle base material.

B. Filler strips. Filler strips shall be examined for softening, swelling, or other deleterious change.

C. Setting compound. The setting compound shall be examined for softening, blistering, cracking, or crazing. After drying the brush, the brush shall be suspended by the handle in a holding device, and a group of bristles, 6 mm. (1/4 in.) in diameter, shall be given a tensile pull of 4.5 kg. (10 lb.) away from the ferrule. Repeat this tensile pull test at random places one time for each inch of brush width.

4.4.3 Drop test. Remove bristle (or synthetic filament) and ferrule. Synthetic handles shall be dropped twice from a height of three meters (10 feet) onto a concrete surface and examined for compliance with 3.1.4.

4.4.4 Straightness. The assembled brush head shall be placed so that the bristle is immersed in 60 degree C (140 degree F) water for five (5) minutes. Excess water shall be shaken out. The brush shall be hanged with bristle down to dry completely. The flare shall be measured by aligning the ferrule portion of the brush as shown in figure 1. Flare measurement is limited to the outline of the main body of the brush. Stray bristles shall be disregarded.

4.4.5 Tipping and flagging of filament. The depth of tipping and flagging shall be measured with a rule and a 7-power magnifying glass.

5. PREPARATION FOR DELIVERY

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5.1 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be in accordance with PPP-B-1880. The level of packaging and packing shall be A, B, C, or otherwise specified in the contract or order (see 6.2). The marking shall be civil, military, or as specified (see 6.2).

6. Notes

6.1. Intended Use. Brushes made with natural hog bristle (Type I) are recommended for application of solvent borne paints and coatings. Brushes made with synthetic filaments (Type II) are recommended for application of water borne paints and coatings.

6.2 Ordering data. Purchasers should select the preferred options permitted herein; and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and size required (see 1).
(c) Levels of packaging and packing required (see 5.1).
(d) Marking required (see 5.1).

6.3 Definitions.

6.3.1 Flagging. Flagging is a mechanical process to split synthetic filament ends to simulate the natural flag of hog bristle.

6.3.2. Tipping. Tipping is a process for grinding or sanding synthetic filament tips to produce a fine, soft, even tip.

PREPARING ACTIVITY

GSA - FSS

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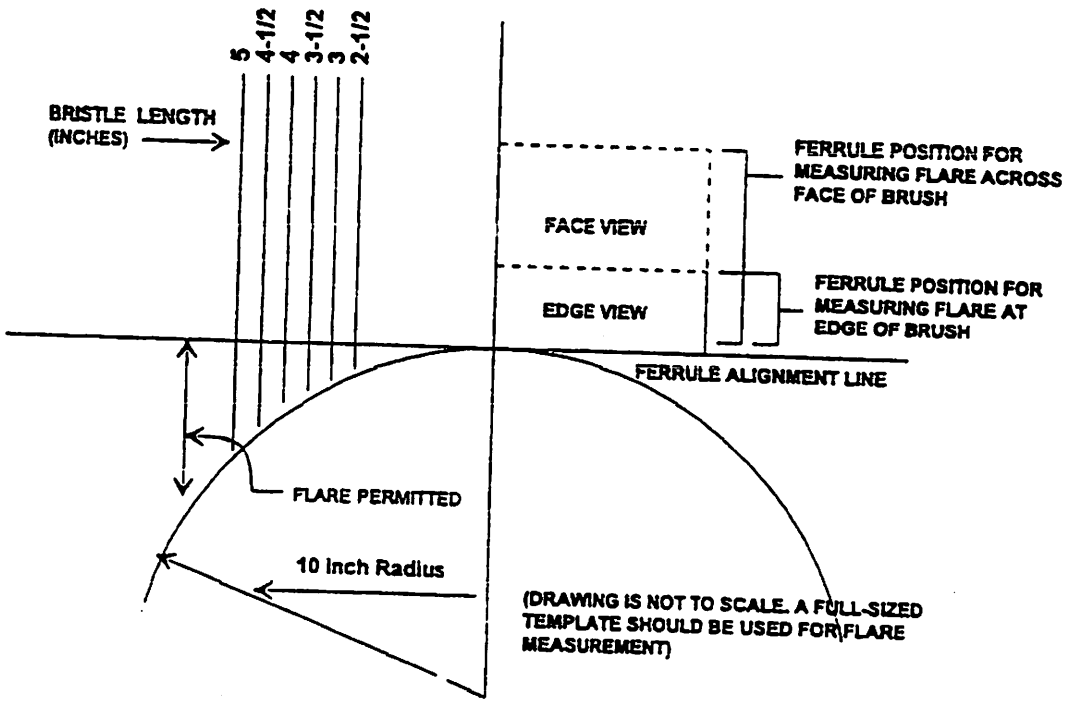


Figure 1. Flare measurement

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